# MAT1100 Homework 2

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**Question 2**

(a) *Proof.* If $n \geq 5$, $A_5$ is simple. Suppose $N \lhd S_n$ then $N \cap A_n \lhd S_n$ and so $N \cap A_n \lhd A_n$. Since $A_n$ is simple this implies that either $N \cap A_n = \{e\}$ or $N \supseteq A_n$.

Case 1. $N \supseteq A_n$. Then $[S_n : N] \leq [S_n : A_n] = 2$ and hence $N = A_n$ or $N = S_n$.

Case 2. $N \cap A_n = \{e\}$, now if $\sigma \in S_n$ then

$$\operatorname{sgn}(\sigma^2) = \operatorname{sgn}(\sigma)^2 = 1$$

hence $\sigma^2 \in A_n$. Therefore every element of $N$ can have order at most 2.

Now since $N$ is normal it is the union of conjugacy classes in $S_n$ hence it is the union of conjugacy classes of the form $(2, \ldots, 2)$ k times for $k \leq n/2$. First note that if $N$ contains the conjugacy class for $k = 1$ then $N$ contains all tranpositions and is therefore all of $S_n$.

First suppose $N$ contains the conjugacy class for $2 \leq k < n/2$. Then $2k < n$. Now pick $(i, j) \in S_n$ and pick $(i_1, j_1) \ldots (i_{k-1}, j_{k-1})$ such that all of these transpositions are disjoint and do not contain either $i$ or $j$. Then since $2k < n$ there exists $\ell \in \{1, \ldots, n\}$ such that $\ell \neq i, j$ and $\ell$ is not contained in any of the above transpositions. Then

$$(i_1, j_1) \ldots (i_{k-1}, j_{k-1})(k, j), (i_1, j_1) \ldots (i_{k-1}, j_{k-1})(i, k) \in N$$

and their product is $(i, j)$ therefore $N$ contains all of the transpositions and is hence of all of $S_n$.

Next suppose $N$ contains the conjugacy class for $k = n/2$. assuming $n$ is even. Pick $i, j, r, s\{1, \ldots, n\}$ all distinct as $n > 4$. We have $k \geq 0$, so let $(i_1 j_1), \ldots, (i_{k-2} j_{k-2}) \in S_n$ be disjoint transpositions such that none of them contain any of $i, j, r, s$. Then

$$(i_1 j_1) \ldots (i_{k-2} j_{k-2})(ir)(sj), (i_1 j_1) \ldots (i_{k-2} j_{k-2})(is)(rj) \in N$$

and their product is

$$(ij)(rs)$$

hence $N$ contains the conjugacy class $(2, 2)$ and by the above case this implies that $N = S_n$. Therefore $N = \{e\}$ if $N \cap A_n = \{e\}$ and the only normal subgroups of $S_n$ are $\{e\}, A_n, S_n$ with quotients $S_n, \mathbb{Z}/2\mathbb{Z}, \{e\}$. □

(b) *Proof.* If $2 \mid |B|$ then by Cauchy's theorem $B$ has an element $x$ of order 2 then there is a map: $S_n \to B$ given by $\sigma \to x$ for $\sigma \in S_n \setminus A_n$ and $\sigma \to e$ for $\sigma \in A_n$. Such a map can be constructed for any element $x \in B$ of order 2. Since $S_n$ has no abelian quotients other than $\{e\}$ and $\mathbb{Z}/2\mathbb{Z}$ these maps and the zero map are all of the possible maps from $S_n$ into $B$. This follows as if $\varphi : S_n \to B$ is a homorphism then

$$S_n/\ker\varphi \cong \operatorname{im} B$$

hence $S_n/\ker\varphi$ must be abelian. □